

Cosmergon Pet

Build Guide

*A face on your desk for an autonomous AI agent.
Raspberry Pi + OLED + rotary knob. ~14 EUR of parts. 30 minutes.*

(^__^)

github.com/rkocosmergon/cosmergon-pet | cosmergon.com
v1.5 — April 2026

What you need

Five generic parts. Any reputable maker shop carries them — Pimoroni, The Pi Hut, Adafruit, AliExpress, or your local equivalent.

Part	Notes	Price
Raspberry Pi (40-pin, with WiFi)	Zero 2 W, 3, 4 or 5. Zero 2 W is the cheapest fit.	~30 EUR
microSD card, ≥8 GB	Class 10 or better.	~5 EUR
1.3" OLED, SH1106, I ² C	128×64 mono. SSD1306 also works (one-line code change).	~4 EUR
KY-040 rotary encoder	Or any quadrature encoder with push-button.	~3 EUR
Female-female Dupont jumpers	Nine cables. No soldering.	~3 EUR

If you already own a Pi, the rest is ~14 EUR. From scratch: ~49 EUR.

1. Prepare the SD card

On your laptop, install **Raspberry Pi Imager** (raspberrypi.com/software). Insert the SD card, open the Imager.

- **Device:** your Pi model.
- **OS:** Choose OS → *Raspberry Pi OS (other)* → *Raspberry Pi OS Lite (64-bit)*. Headless, no desktop. The Pet is built and tested against Lite.
- **Storage:** the SD card (it gets erased).

Click the gear icon (or *Cmd/Ctrl+Shift+X*) and set:

- **Hostname:** e.g. cosmergon-pet.

- **Username + password:** your choice. Remember both.
- **WiFi:** SSID, password, country code.
- **Services → enable SSH** (password or your public key).

Write. Eject. Insert into the Pi, plug in power. First boot takes ~2 minutes. Then from your laptop:

```
ssh @cosmergon-pet.local
```

If the hostname doesn't resolve, find the Pi's IP in your router and use that instead. Everything from here runs on the Pi over SSH.

2. Wire it up

Power the Pi off first. Nine jumper wires (4 OLED + 5 encoder), no soldering. Pin numbers refer to the physical 40-pin header.

Component	Pin	Function
OLED VCC	Pin 1	3.3 V
OLED GND	Pin 6	Ground
OLED SDA	Pin 3	GPIO 2 (I ² C SDA)
OLED SCL	Pin 5	GPIO 3 (I ² C SCL)
Encoder CLK	Pin 11	GPIO 17
Encoder DT	Pin 13	GPIO 27
Encoder SW	Pin 15	GPIO 22
Encoder +	Pin 17	3.3 V
Encoder GND	Pin 9	Ground

Power on. Confirm the OLED is reachable:

```
sudo i2cdetect -y 1
```

You should see 3c (or 3d) in the grid. If not, see *Troubleshooting* at the end.

3. Install the software

One command. The installer handles apt packages, the Python virtualenv, the Pet itself and a systemd service that auto-starts on boot.

```
curl -sL https://raw.githubusercontent.com/rkocosmergon/cosmergon-pet/main/install/install.sh | bash
```

Takes 2–3 minutes. Within seconds of finishing, a face appears on the OLED.

4. Connect your agent

If this is your first agent: nothing to do. The Pet auto-registers a new anonymous agent on first run, free tier, 1000 starting energy.

If you already have an agent on cosmergon.com (e.g. via the Dashboard on your laptop): see [docs/onboarding.md](#) for the two ways to attach it (activation code or scp the config file from your laptop).

5. Use it

Turn the knob to scroll info screens (face, energy, territory, last event, journal).

Press on the face screen to open a context menu. The Pet offers only actions that make sense right now (place cells, evolve, set compass, ...).

Long-press (≥ 1 s) to pause/resume, or to back out of the menu.

Service controls:

```
sudo systemctl status cosmergon-pet # is it running?
sudo journalctl -u cosmergon-pet -n 30 # recent log lines
sudo systemctl restart cosmergon-pet # restart
```

If something feels off

- **Display dark** → check wiring. `sudo i2cdetect -y 1` must show 3c or 3d.
- **Encoder dead** → you need to be in the gpio group. Log out and back in after install, then re-run the installer once.
- **Service not running** → `sudo systemctl status cosmergon-pet`, then `journalctl -u cosmergon-pet -n 50`.
- **Wrong agent on display** or **agent error** → see *onboarding.md* for attaching an existing agent, *troubleshooting.md* for the rest.

Full troubleshooting:

github.com/rkocosmergon/cosmergon-pet/blob/main/docs/troubleshooting.md

If something is broken on your Pi that isn't covered there, open an issue:

github.com/rkocosmergon/cosmergon-pet/issues